

Lower Cretaceous radiolarian bedded chert from the Mineoka Belt, Boso Peninsula, Japan

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Abstract: A meter-scale, bedded red radiolarian chert block was found on the Yo-oka Beach at the eastern edge of the Mineoka Belt, Boso Peninsula, central Japan. The block is included in the area composed of fragments of the Mineoka ophiolite and other sedimentary and igneous rocks, including a Paleocene to Miocene pelagic limestone/chert sequence. The radiolarian fauna indicates an assemblage zone of middle to late Albian (Early Cretaceous), partly corresponding to the age of the Tethyan fauna. The tectonic significance of this Cretaceous radiolarian chert block is not clear: it could be from either the Mineoka ophiolite and related oceanic plate stratigraphy or from the Shimanto Supergroup.

Key words: Mineoka Belt, ophiolite, oceanic plate stratigraphy, Albian (Lower Cretaceous), radiolarian chert

Introduction

The Mineoka Belt is a complex fault zone composed of ophiolitic rocks and pelagic, hemipelagic and terrigenous sedimentary rocks, running from east-west in the southern Boso Peninsula, central Japan (Fig. 1) (Ogawa and Taniguchi, 1987, 1988; Sato et al., 1999; Hirano et al., 2003; Takahashi et al., 2003; Ogawa and Takahashi, 2004). The ages of the basaltic rocks are dated as either Eocene or Oligocene, ranging from 40 to 50 Ma by Ar-Ar radiometric dating (Hirano et al., 2003), and those of the pelagic limestone/chert sequence range from the late Paleocene to middle Miocene through Eocene and Oligocene (Mohiuddin and Ogawa, 1996, 1998a, b). Above all the oldest sedimentary rocks are known to be the upper Paleocene limestone in Nishi (Mohiuddin and Ogawa, 1998b), 10 km west from the Yo-oka Beach where the presently reported Cretaceous chert was found, although some of the exact ages are still under debate. Ogawa and Taniguchi (1988) and Mohiuddin and Ogawa (1998b) considered that the Paleocene to Miocene pelagic limestone/chert sequence might overlie the subjacent

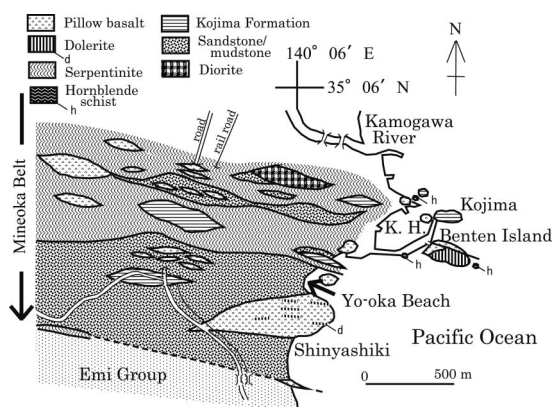


Fig. 1. Simplified map showing distribution of rocks in the eastern Mineoka Belt, central Japan. (Modified from Takahashi et al. (2003)). K. H. means Kamogawa harbor.

oceanic plate rocks, which are in part among the Mineoka ophiolite. Ogawa and Taniguchi (1988) suggest such oceanic rocks to be of the “Mineoka plate”, and Sato et al. (1999) support this idea from various petrochemical and tectonic aspects.

However, the problem still remains unsolved as to why the basaltic rocks of the ophiolite are younger than the oldest pelagic rocks, although there is some possibility that the oldest part of the Mineoka plate might be older than the presently remaining basaltic rocks, and that only the younger basalts and the oldest pelagic rocks are present in the Mineoka Belt. Hirano and Okuzawa (2002) and Hirano et al. (2003) report a terrigenous sandstone which is included within alkali basaltic pillow lava of ca. 20 Ma, and interpret that the alkali basalt erupted around the trench area during the last stage of the Mineoka plate.

Other than the Mineoka plate and the related seamount and sedimentary rocks in the Mineoka Belt, there are some sedimentary rocks which are correlated to the Shimanto Supergroup (Nakajima et al., 1981). They are sandstone and shale that are not very well dated, except for middle Eocene radiolarians from a calcareous nodule in the shale (Kawakami, 2004).

We found a Lower Cretaceous (as explained later as middle to late Albian) radiolarian fossil-bearing bedded red chert inside of the Mineoka Belt. This rock is float on Yo-oka Beach, at the easternmost edge of the belt, south

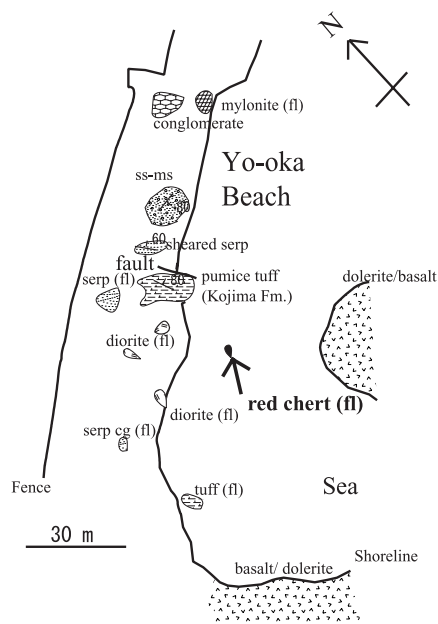


Fig. 2. Outcrop sketch map of Yo-oka Beach. (fl) means float.

of the Kamogawa harbor, but is considered reasonably to be from nearby (Figs. 1, 2).

Occurrence of bedded chert

This chert block is a meter diameter boulder on the sand beach (Fig. 3). Around this area there are several rocks of basalt, serpentinite, pumiceous tuff (not so highly deformed, probably part of the middle Miocene Kojima Formation (Ogawa and Taniguchi, 1988), sandstone and conglomerate (not so highly deformed, probably part of the middle Miocene Sakuma Group (Inoue and Nakaseko, 1951; Koike, 1957), because foraminifers indicative of Blow's Zone N8 were found by M. M. Mohiuddin (personal communication, 2003) as outcrops) (Fig. 2). As far as exposed, these rocks are in fault contact with each other. On the beach there are many blocks (mostly boulders) of basalt, gabbro and other ophiolitic rocks as float, but no other chert blocks are seen than those reported in this paper.

To the north and south of Yo-oka Beach, several huge (some 100 m-size) blocks of tholeiitic and alkaline basalt are distributed (Takahashi et al., 2003; Hirano et al., 2003). The tholeiitic basalts are either mid-oceanic ridge basalt (MORB), back-arc basin basalt (BABB) or island arc tholeiite (IAT) in chemistry, and the alkaline basalt is within-plate basalt (WPB) (Hirano et al., 2003), suggesting that those rocks are composed of part of oceanic plate or part of an island arc, including a possible back-arc basin setting, with a hotspot type oceanic island on top.

The present chert is reddish, bedded, and isoclinally folded (Fig. 3a). Thin section observation indicated that

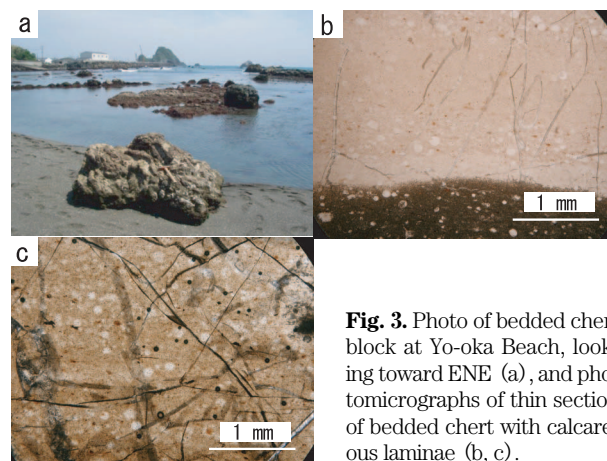


Fig. 3. Photo of bedded chert block at Yo-oka Beach, looking toward ENE (a), and photomicrographs of thin section of bedded chert with calcareous laminae (b, c).

thinly laminated or bedded siliceous and calcareous alternation is the original sedimentary structure (Fig. 3b, 3c). Most, but not all, radiolarians are moderately preserved. Four samples (YO030418-4-1 to 4) were collected from the block, and were dissolved in diluted HF liquid, and extracted radiolarian tests were studied by using SEM photos (Fig. 4).

Radiolarian age

All the four samples contain almost the same radiolarian assemblage. Sample YO030418-4-4, which is best preserved among the four, yields the following species: namely, *Tugurium pagoda* (Squinabol), *Parvicingula* sp., *Parvicingula?* sp., *Dictyomitra formosa* Squinabol, *Histocapsa* sp., *Dorypyle?* sp., *Praeconosphaera* sp., *Archaeodictyomitra* sp., *Xitus* cfr. *spinosus* (Squinabol), *Rhopalosyringium* sp., *Rhopalosyringium?* sp., *Acaeniotyle* cfr. *diaphorogona* Foreman, *Sciadiocapsa speciosa* (Squinabol), *Acanthocircus levis* (Donofrio and Mostler), *Crucella* sp., *Pseudodictyomitra?* sp. and others.

Most of these radiolarians have been reported from middle Cretaceous pelagic and hemipelagic sequences in the Mediterranean regions by O'Dogherty (1994). The co-occurrence of *Tugurium pagoda*, *Dictyomitra formosa*, *Sciadiocapsa speciosa*, and *Acanthocircus levis* indicates that the age of this radiolarian assemblage is middle to late Albian, late Early Cretaceous (Figs. 4, 5).

To date, upper Lower Cretaceous radiolarians from the North Pacific have been studied by Foreman (1975) (DSDP Leg 32) and by Schaaf (1981) (DSDP Leg 62). Foreman (1975) sets up the *Acaeniotyle umbilicata* zone for core samples of pelagic clay and chert of the North Pacific (the region within 28° to 40° N, 154° to 161° E). However, there is only one related species, *Acaeniotyle* cfr. *diaphorogona* and *Acaeniotyle diaphorogona* in both the present Mineoka samples and

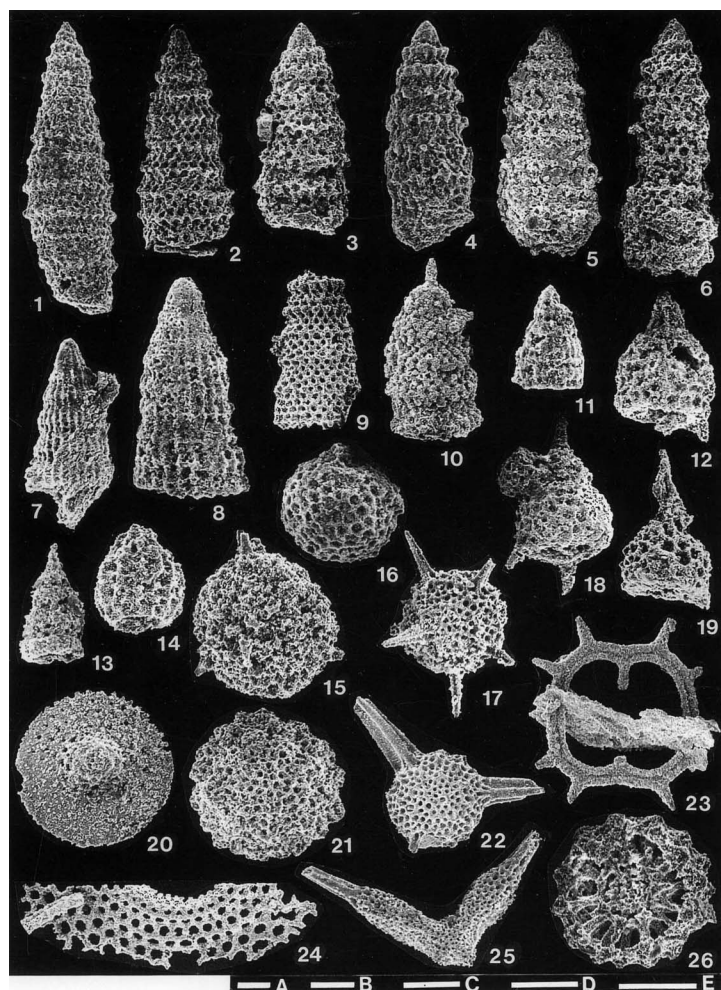


Fig. 4. Scanning electron micrographs of selected radiolarians from sample YO030418-4-4. Each species and magnification of test is as follows. Scale bars A to E indicate 50 μ m. A applies to 20, B to 1, 2, 4, 5, 8, 9, 22-25, C to 3, 6, 7, 10, 13, 15, 16, 18, 19, 21, D to 11, 12, 26, E to 14, 17.

1. *Tugurium pagoda* (Squinabol) 2. to 6. *Parvicingula* sp. 7. *Dictyomitra formosa* Squinabol 8. *Archaeodictyomitra*? sp. 9. *Parvicingula*? sp. 10. *Xitus* cfr. *spinus* (Squinabol) 11. *Archaeodictyomitra*? sp. 12. *Rhopalosyringium* sp. 13. *Pseudodictyomitra*? sp. 14. *Rhopalosyringium*? sp. 15. *Dorypyle*? sp. 16. *Hiscocapsa* sp. 17. *Staurosphaeretta* sp. 18. and 19. *Rhopalosyringium* sp. 20. *Sciadiocapsa speciosa* (Squinabol) 21. *Praeconsphaera* sp. 22. *Acaeniotyle* cfr. *diaphorogona* Foreman 23. *Acanthocircus levis* (Donofrio and Mostler) 24. *Sciadiocapsa* sp. 25. *Crucella* sp. 26. *Praeconsphaera* sp.

those of DSDP Leg 32, respectively. Schaaf (1981) also introduced the *Acaeniotyle umbilicata* zone for lower Aptian to middle Albian samples of the North Pacific (the region within 21° to 39° N, 173° to 179° E). Only *Acaeniotyle* cfr. *diaphorogona* and *Acaeniotyle diaphorogona* are also the related species between the Mineoka and DSDP Leg 62 area. Some of the species are the same as those from chert blocks within the Shimanto Supergroup in Shikoku (e.g., Taira et al., 1988), although not all the species of the present assemblage correspond to those from the Shimanto Supergroup. As a result, in general, these species suggest that this rock is correlatable to the middle to upper Albian (Lower Cretaceous) (Fig. 4), but the exact correlation between the species of several surrounding places is still not clear.

Discussion

The present report is the first finding of Cretaceous fossils from the Mineoka Belt. These fossils are older than the ages previously known from the pelagic sedimentary rocks and basaltic rocks. The problem is what kind of tectonic significance this rock has, and what the relation to

the ophiolitic rock is. One possibility is that this rock is the part of the Shimanto Supergroup, which is distributed from Kyushu to the Kanto Mountains through Shikoku, Kii and the Akaishi areas. There is known in the Mineoka Belt a turbidite sequence of similar lithology to the Shimanto Supergroup, the Hatcho Formation (Nakajima et al., 1981). However, this formation is thought to be of Paleogene age because of lithological similarity to the Paleogene Shimanto in the Setogawa Group (Nakajima et al., 1981), and by its middle Eocene radiolarian fauna (Kawakami, 2004). No Cretaceous fossils have ever been reported from the Hatcho Formation.

Another possibility is that the present chert is part of the pelagic sedimentary cover above oceanic plate rocks, that are almost equivalent to the Mineoka ophiolitic rocks. There is no information on the other similar examples of bedded red chert of similar age from the Mineoka Belt or from nearby areas. The closest known examples are from Ocean Drilling Program (ODP) Site 808 in the NW Pacific, and also from a seamount on the Pacific side of the Mariana Trench. From the latter, the lowest Cretaceous to upper Cretaceous radiolarian-bearing mudstone and red

	Early Cretaceous		Late Cretaceous	
	Albian		Cenomanian	Turonian
Radiolarians				
<i>Tugurium pagoda</i>	—			
<i>Dictyonitra formosa</i>				
<i>Sciadiocapsa speciosa</i>	—			
<i>Acanthocircus levis</i>	—			

Fig. 5. Known ranges of selected Cretaceous radiolarians from the present samples, Mineoka Belt. Base data are from De Wever et al. (2001).

laminated cherts were collected by a submersible from the flank of the seamount, which is collapsing into the Mariana trench (Ogawa et al., 1996; Ogawa and Kawata, 1998). Those rocks are not all silicified but one is an unce-mented mudstone, some are between chert and porcel- lanite. Some parts of those rocks resemble the lithology of the present chert, as both have calcareous laminae.

There is some differences in lithology between the ordi- nary radiolarian cherts from the Shimanto Supergroup, such as that from the Tei mélange in Shikoku (Taira et al., 1980), which is apparently originates from the Kula- Pacific plate, and the examples from the Mineoka Belt reported in this paper and those from the Mariana trench area in the present Pacific plate as mentioned above. It is still not clear from what part of the plate our present example came. We need further discussion on the paleo- geographic and tectonic implications of this present chert block.

Summary and conclusion

The bedded red chert block from Yo-oka Beach, Mineo- ka Belt, bears Early Cretaceous (middle to late Albian) radiolarians. The tectonic significance of this block depends on whether it originates from the Shimanto Supergroup or from the oceanic plate from the Pacific side.

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* in Japanese with English abstract ** in Japanese

用語対比

Mineoka Belt	嶺岡帯	Kamogawa	鴨川
Kojima Formation	荒島層	Sakuma Group	佐久間層群
Yo-oka Beach	八岡海岸		